

Pilot experiments

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Predefining an analysis plan

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Research ethics and consent

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Ablation studies

Without opening the old lesson, explain one key idea from Ablation studies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Pilot experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Pilot experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Pilot six cases exposes ambiguous labels in two. What should happen before the primary study?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Pilot five cases reveals wrong units in duration logs; team corrects units and reports the pilot as final evidence. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A pilot may assess feasibility but is not automatically powered or independent evidence for the final hypothesis. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Pilot six cases exposes ambiguous labels in two. What should happen before the primary study?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled pilot experiments evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of pilot experiments. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

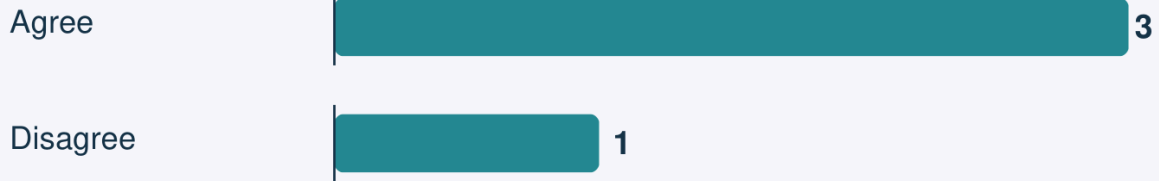
12. Pictorial case practice: Pilot experiments

A pilot tests the procedure. A pilot checks whether the team can score answers consistently before running the main study.

Two reviewers score four practice answers. They agree on three and disagree on one; these answers are excluded from final evaluation.

Reviewer agreement on four pilot answers

pilot answers out of 4



Bar length shows the supplied value; the marked baseline is zero.

Pilot experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Try the workflow

Both reviewers apply the draft rubric to four practice answers.



2. Measure agreement

Agreement is $3/4=75\%$ on this pilot.



3. Repair the rubric

Discuss the disagreement and clarify the scoring rule before the main run.

CHANGE ONE THING

Add the corrected pilot answers into final evaluation.

Explain the changed result using the matching lesson.

Calculate reviewer agreement on the four pilot answers. What can the disagreement improve, and why are those revised practice answers not untouched final evidence?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A pilot tests the procedure. A pilot checks whether the team can score answers consistently before running the main study.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Discuss the disagreement and clarify the scoring rule before the main run.

1. Try the workflow
2. Repair the rubric
3. Measure agreement

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Repair the rubric Discuss the disagreement and clarify the scoring rule before the main run. A pilot can reveal missing instructions, practical delays or ambiguous scoring. It is preparation, not an untouched final estimate after the team uses it to revise methods.

Practice 2: explain the deciding evidence

Changed-case practice: Add the corrected pilot answers into final evaluation. Which conclusion is supported by the supplied facts?

1. Declare them previously exposed rather than untouched final evidence.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Declare them previously exposed rather than untouched final evidence. Rubric development already used those answers.

Repair a missing building block

Review Predefining an analysis plan (ai-7/21). Goal: Explain the mechanism and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the mechanism and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes. Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Review Ablation studies (ai-7/16). Goal: Explain the mechanism and terminology of ablation studies. Short practice: Calculate the filter ablation's two rates and count change. Which single component is altered, and why does also changing the model prevent isolating the filter?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

